Check for updates

Received: 5 November 2021 | Revised: 8 January 2022 | Accepted: 22 January 2022

IET Image Processing

DOI: 10.1049/ipr2.12443

The Institution of Engineering and Technology WILEY

ORIGINAL RESEARCH PAPER

Perceptual redundancy model for compression of screen content videos

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Funding information
National Natural Science Foundation of China, Grant/Award Number: 61871437; Natural Science Foundation of Hubei Province, Grant/Award Number: 2019CFA022

Abstract

Screen content video (SCV) consists primarily of text areas, computer graphics and other computer-generated content and possesses unique perceptual characteristics. To compress SCVs more effectively with less reduction in subjective quality, perceptual characteristics of SCVs are analyzed and a perceptual redundancy (PR) model for SCV compression is proposed, including spatial PR (SPR), temporal PR (TPR) and foveated PR (FPR) model. In SPR modeling, the SCV is divided into sharp edge (SE) areas and non-SE areas, then SPR is estimated separately. In TPR modeling, both inter-frame luminance adaptation effect and motion masking effect are taken into account. In FPR modeling, each frame of SCV is classified into abrupt frames, relative motion frames or static frames. Then fixation points of different kinds of frames are predicted using different methods, and FPR is modeled considering foveated masking effect and visual attention. Finally, the perceptual redundancy of SCV is estimated based on the product of SPR, TPR and FPR. It is experimentally demonstrated that compared to the state-of-the-art models, the authors' model could obtain more accurate estimates of PR. Moreover, the model is incorporated into SCV compression with an adaptive perceptual quantizer. An average of 7.42% bits could be saved with less decline in subjective quality.

1 | INTRODUCTION

With the rapid development of network communication and multimedia technology, applications of video coding are becoming more and more diversified. Recently, online education and e-commerce have become industry hotspots, driving an explosion in demand for online teaching, webcasting and online conferences. These applications share the same characteristic, that is, screen content videos (SCVs) need to be compressed and transmitted with high compression efficiency. From the perspective of user experience, SCVs should be compressed with the best perceptual quality under bandwidth constraints, which puts forward higher requirements for SCVs compression.

Different from the video captured directly by the camera, SCVs are mainly composed of computer-generated contents, such as text, vector graphics, software icons and cursors, or a mixture of the content captured by the camera and computer-generated content. The compositional differences result in unique spatio-temporal structural characteristics, for instance, strong contrast, high color purity etc. In order to compress SCVs more efficiently, the Joint Collaborative Team on Video coding (JCT-VC) has developed an extended tool set for screen content based on high efficiency video coding (HEVC) [1], which is called HEVC extensions on Screen Content Coding (HEVC-SCC) [2]. This extension tool kit takes into account the distinctive features of SCVs, and several coding tools [3, 4] have been added to reduce redundancy.

From the perspective of perception, the human visual system (HVS) could not perceive the coding distortion below a certain threshold, which is called the just noticeable distortion (JND). It is usually used to represent the perceptual redundancy (PR) in images or videos, and reflects the sensitivity of HVS to compression distortion of videos [5]. Thus, the JND estimation has been widely incorporated into the video compression, and the PR-based perceptual video coding (PVC) has been greatly developed. In general, existing PR models can be divided into two main categories, that is, pixel-based PR models and subband-based PR models. The pixel-based PR model is built in the pixel domain, which is applied to estimate the PR threshold of each pixel [6–13]. In terms of the subband-based PR model,

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it is usually constructed in the frequency domain (e.g., DCT and DWT) to estimate the perceptual redundancy for each transform coefficient [14–21]. Recently, some researches break the limitations of subband-based PR models and pixel-based models, and try to model PR from other perspectives, such as patch, picture and video-video PR model [22–25].

However, PR models mentioned above were built for natural images or natural videos. Due to the unique spatial-temporal structure characteristics of SCVs, the perceptual redundancy of SCVs is more likely to be overestimated, especially in the sharp edge areas of SCVs. In order to avoid the overestimation problem of perceptual redundancy, several PR models for screen content images (SCIs) have been proposed. In [26], Wang et al. decomposes the edge profile into luminance, contrast and structure, and estimate sub-PR thresholds for each component in different ways. The perceptual redundancy of SCIs is finally expressed by the non-linear superposition of three sub-PR thresholds. In [27], Liu et al. decomposes the SCI into screen content set (SCS) and the non-SCS. For the SCS, the perceptual redundancy is estimated with Wang's model [26]. As for the non-SCS, the perceptual redundancy is estimated based on the edge masking effect and the pattern masking effect. In [28], Wang et al. considers high-frequency and orientation sensitivity of HVS to further estimate the perceptual redundancy of regular contours and strong contrast of SCIs. However, these pixel-based PR models are developed for the static SCI, and they may not be accurate in predicting perceptual redundancy of SCVs due to the following drawbacks. 1) HVS is sensitive to sharp edges of SCVs but not to weak edges. The edge areas of SCVs should be divided into the sharp edge (SE) areas and the non-SE areas, and the perceptual redundancy of SE and non-SE should be estimated differently to improve the accuracy of PR model. 2) For the SCV, the temporal masking effect should be introduced into PR estimation. 3) During video playback, HVS only pays attention to visual information in a limited number of fixation areas, and ignores that in non-fixation areas. Thus, the foveated masking effect should be appropriately considered when estimating PR for SCVs.

In order that the perceptual redundancy for SCVs could be estimated more accurately, we propose a novel foveated PR estimation model for SCVs, which consists of the spatial PR (SPR) model, the temporal PR (TPR) model and the foveated PR (FPR) model. In the SPR modeling, the sharp edge areas will be detected and its perceptual redundancy will be estimated based on the estimation of blurry distortion. In the TPR modeling, we use an information-theoretic approach to analyze the temporal masking effects caused by inter-frame differences and motion. Then the TPR model is built as an scaling factor of the SPR model. In the FPR modeling, video frames of SCVs are divided into the abrupt frame, the relative motion frame and the static frame according to a proposed frame category classifier. Then different methods are used to predict the fixation points of each type of frames. The foveated PR model is built as another scaling factor considering the visual attention and the foveated masking effect. Finally, our SCV-based PR model is represented as the product of the SPR model, the TPR model and the FPR model.

The main contributions of this paper are briefly as follows: (1) The sharp edge areas of SCVs are detected and an enhanced spatial PR model is proposed for SCVs. (2) The temporal PR model is proposed for PR estimation of SCVs for the first time. (3) A frame category classifier is designed to describe the inter-frame change of SCVs, and a foveated PR model is proposed for PR estimation for SCVs. Experimental results show that our model performs best in terms of distortion masking capability compared to other state-of-the-art PR models. Besides, our PR model is incorporated into SCV compression in the manner of perceptual quantization. Experimental results of SCV compression show that our model could save over 7% of bit rate with the least subjective quality decline compared to other state-of-the-art PR models.

The remainder of this paper is organized as follows: Section 2 proposes a frame category classifier for SCVs based on the temporal characteristics of SCVs. The framework of the proposed SCV-based PR estimation model is then described. In Section 3, we introduce the modeling process of the SPR model and the TPR model. Section 4 introduces the fixation prediction of SCVs and presents the modeling process of the FPR model. Sections 5 and 6 show the experimental results and conclude this paper, respectively. Table 1 lists the abbreviations and corresponding explanations of this article.

2 | THE FRAMEWORK OF THE SCV-BASED PR MODEL

As mentioned in Section 1, the SCV is different from the natural video in spatial-temporal structure, and possesses unique perceptual characteristics of SCVs. Therefore, in this section, we first introduce the spatial-temporal structure characteristics of SCVs, then, based on the differences between adjacent frames of SCVs, we propose a frame category classifier for the first time. Finally, the framework of the proposed SCV-based PR model is introduced.

2.1 | Characteristics of screen content videos

2.1.1 | Spatial characteristics of SCVs

Computer-generated content, such as text, vector graphics, software icons, cursors, appears more frequently in SCVs. This results in sharper edges and larger contrast, especially when natural regions are not included. Besides, due to the noise-free property of computer generated content, detailed textures are relatively more regular, and local areas may have identical visual content, or be concentrated around several colors.

2.1.2 | Temporal characteristics of SCVs

As shown in Figure 1, we provide several situations about inter-frame motion among SCVs, including scene switching, relative motion and static motion. According to the motion

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TABLE 1 Abbreviations and explanations in this article

Abbreviations	Explanations
SCV	Screen Content Video
SCI	Screen Content Image
HVS	Human Visual System
JND	Just Noticeable Distortion
PR	Perceptual Redundancy
PVC	Perceptual Video Coding
DCT	Discrete Cosine Transform
DWT	Discrete Wavelet Transform
SCS	Screen Content Set
SE	Sharp Edge
FC	Frame Category
SPR	Spatial PR
TPR	Temporal PR
FPR	Foveated PR
LA	Luminance Adaptation
ECM	Edge Contrast Masking
TCM	Texture Contrast Masking
MM	Motion Masking
FM	Foveated Masking
AB-frame	Abrupt Frame in SCVs
RM-frame	Relative Motion Frame in SCVs
ST-frame	Static Frame in SCVs
JNB	Just Noticeable Blur
PSNR	Peak Signal to Noise Ratio
MOS	Mean Opinion Score
DMOS	Differential Mean Opinion Score
DSCQS	Double Stimulus Continuous Quality Scale
QP	Quantization Parameter
CU	Coding Unit

of two adjacent frames, the video frame can be classified into the abrupt frame (AB-frame), the relative motion frame (RM-frame) and the static frame (ST-frame). The AB-frame is mainly caused by scene switching, such as PPT playing, window closing and web page opening. Under these circumstances, motion and visual content in the AB-frame are unpredictable, resulting into the stronger temporal masking effect. In terms of the RM-frame and the ST-frame, the visual content can be well predicted. The fixation areas will be detected quickly, which will enhance the visual perception of the SCVs. In general, the probability of AB-frames and ST-frames in SCVs is significantly higher compared to natural videos. Therefore, it is necessary to analyze and classify the motion in SCVs.

2.1.3 | Perceptual characteristics of SCVs

When perceiving natural videos, observers perceive them mainly by comparing the foreground and background of the image, or the relative position, color difference, luminance or contrast difference between objects. However, SCVs are mainly composed of computer-generated contents, such as text, vector graphics, software icons and cursors. In order to extract more information from these high-level semantic contents, the HVS will pay more attention to these contents. As a result, distortions in these areas are relatively more easily to be perceived.

2.2 | Frame category classifier for screen content videos

Motion characteristics of different types of screen content frames are quite different, which leads to different visual characteristics. Hence, it is necessary to design a frame category classifier for SCVs to better fit the features of HVS using different PR modeling schemes. In this article, we propose a motion-based frame category classifier, which classifies screen content frames into AB-frames, RM-frames and ST-frames based on the motion information entropy.



(a) Scene switching (b) Relative motion of large area

(c) Relative motion of local area (cursor movement) (d) Adjacent frames without any differences (static movement)

FIGURE 1 Several situations of inter-frame motion in SCVs

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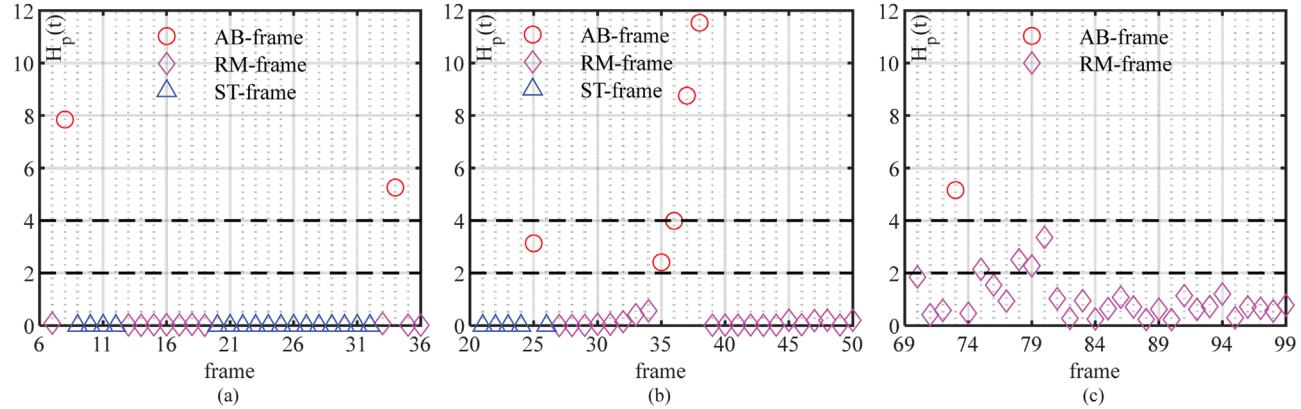


FIGURE 2 H_p of AB-frames, RM-frames and ST-frames: (a) WebBrowsing; (b) WordEditing; (c) Programming

Specifically, we use optical flow [29] to estimate the motion vector $\vec{v}$ of each pixel of the current frame with respect to the previous frame, $\vec{v} = (v_x, v_y)$. The frequency of each motion vector is represented by $p(\vec{v})$ and its information entropy $H_p(t)$ for t^{th} frame is calculated as follows:

$$H_p(t) = -\sum_{v_x=-\infty}^{+\infty}\sum_{v_y=-\infty}^{+\infty} p(\vec{v})\log_2 p(\vec{v}). \tag{1}$$

Figure 2 shows the distribution of $H_p(t)$ in the 'WebBrowing', 'WordEditing' and 'Programming' sequences. Due to the obvious and irregular motion, the probability distribution of motion vectors of AB-frames tends to be relatively uniform, resulting in relatively larger motion information entropy. Motion information entropy is mostly distributed in the region of $H_p(t) > 4$ for AB-frames. In addition, this figure also reflects that the distribution of AB-frames in SCVs is relatively scattered. In other words, AB-frame exist in the form of a single frame or several continuous frames. For example, the 35th frame and the next three frames of the sequence 'WordEditing' are AB-frames, where a window is opening. There are two situations in RM-frames: one is when motion is concentrated in a small area, and the motion information entropy is relatively small, concentrated in the region of $H_p(t) < 2$, and the other occasion is when motion is more intense, but is more regular and may last for a while, in this occasion, $H_p(t)$ is mostly within the interval of 2 to 4, and motion information entropy of adjacent frames fluctuates little. For example, in the 73rd frame of the sequence 'Programming', there is a window pop-up, which means it is an AB-frame, and in the subsequent frames, there is a large relative motion that causes the motion information entropy to fluctuate in the range of 2 to 4, but the fluctuation range is small, and these frames are RM-frames. The motion vector of ST-frame is always zero, and motion information entropy is always zero.

In order to better distinguish between AB-frames and RM-frames with large relative motion area, it is necessary to consider the changing pattern of motion information entropy over a period of time. We first propose a perceptual sliding window l_{sw} to characterize this time period, whose length is calculated by the equation (2), indicating the number of frames in the sliding window.

$$l_{sw} = \lfloor f \times t_{sr} \rfloor, \tag{2}$$

where t_{sr} is the visual simple reaction time, which is used to estimate the reaction latency of HVS, and is set to 199 milliseconds [30]. Note that when the duration of relative motion is longer than t_{sr}, HVS can react to the changes, so as to distinguish from AB-frames. f is the frame rate of the current sequence. $\lfloor\cdot\rfloor$ represents the rounding down operator.

The average motion information entropy of the sliding window is computed by $H_a = \sum_{t_{sr}} H_p(t)/t_{sr}$. Finally, the frame category (FC) classifier Ψ could be described by:

$$\Psi(t) = \begin{cases} ST-frame, & H_p(t) = 0, \\ RM-frame, & H_p(t) \in K, \\ AB-frame, & else, \end{cases} \tag{3}$$

where K represents $\{0 < H_p(t) \le 2\} \cup \{2 < H_p(t) < 4 \,\&\, H_p(t) < \xi \times H_a\}$, in which ξ is a constant and set to 3 [31].

2.3 | The framework of PR model for screen content videos

In order to better adapt to the characteristics of HVS when viewing SCVs, a novel PR estimation model is proposed for SCVs according to the frame category classifier, and its framework is shown in Figure 3. As shown in this figure, the proposed SCV-based PR model consists of the SPR model, the TPR model and the FPR model. In SPR modeling, the current frame is divided into the SE areas and the non-SE areas according to sharp edge detection. In the SE areas, the SPR model will be built based on the luminance adaptation (LA) effect and the edge contrast masking (ECM) effect. As for the non-SE areas, its SPR model will be built based on the LA effect and the texture contrast masking (TCM) effect. In TPR modeling, we will compute the inter-frame residue and

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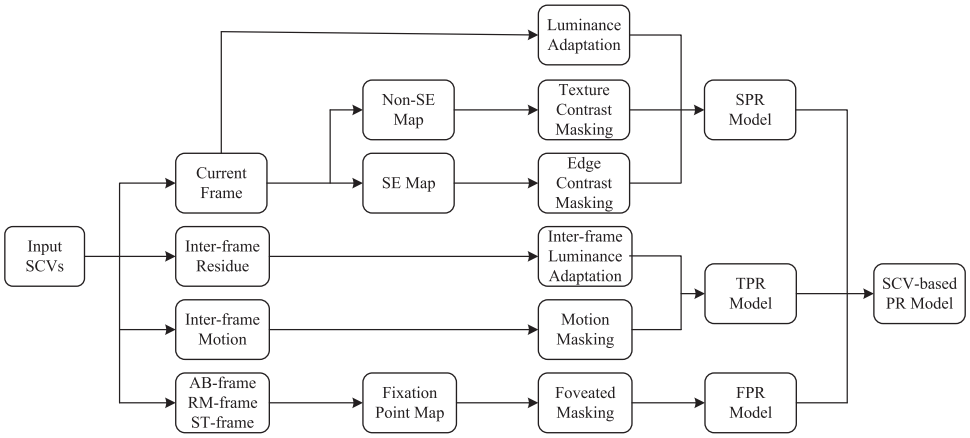


FIGURE 3 The framework of PR estimation for SCVs

the inter-frame motion. Then, the TPR model is established as a scaling factor according to the inter-frame LA effect and the motion masking (MM) effect. In the FPR modeling, the current frame will be classified and its fixation points will be predicted. The FPR model is then built as another scaling factor according to the foveated masking (FM) effect. Finally, the SCV-based PR model is defined as the product of the SPR, TPR and FPR models.

3 | SPATIAL-TEMPORAL PR ESTIMATION FOR SCVs

As mentioned above, SCVs possess unique spatial and temporal characteristics. To better fit the spatio-temporal features when perceiving SCVs, in this section, we will introduce the SCV-based spatio-temporal PR model. In particular, in the SPR model, sharp edges in SCVs are separated according to the edge intensity, and then we estimate the contrast masking effect in sharp edge and non-sharp edges areas using different methods. In the TPR model, in addition to inter-frame luminance adaptation effect, we use a sigmoid function to represent the effect of the motion masking effect.

3.1 | Sharp edge detection of SCVs

Sharp edges of screen content images are mainly concentrated at the verge between text and background, boundaries of computer-generated graphics, and foreground-background junction in natural areas. Due to the non-noise characteristic of SCVs, it is not necessary to use Gaussian smoothing filter to reduce the noise. In this paper, the sobel operator is used for edge detection because of its excellent anti-interference. For the pixel z, its horizontal and vertical gradients $grad_x$, $grad_y$ could be calculated as follows:

$$grad_x(z) = I(z) * G_x,\ grad_y(z) = I(z) * G_y. \tag{4}$$

Here, $I(z)$ is the luminance of the pixel z, and $*$ is the convolution operator. G_x and G_y are expressed by

$$G_x = \begin{bmatrix} -3 & 0 & 3 \\ -10 & 0 & 10 \\ -3 & 0 & 3 \end{bmatrix}, \quad G_y = \begin{bmatrix} -3 & -10 & -3 \\ 0 & 0 & 0 \\ 3 & 10 & 3 \end{bmatrix}. \tag{5}$$

Finally, the gradient of pixel z is expressed with $grad(z) = \sqrt{grad_x(z)^2 + grad_y(z)^2}$.

Obviously, the gradient of sharp edge areas is relatively higher. Hence, a threshold θ_{th} needs to be set to filter out sharp edges. According to this threshold, the pixel z could be divided into the SE pixel and the non-SE pixel as follows:

$$SE(z) = \begin{cases} 1, & grad(z) > \theta_{th}, \\ 0, & grad(z) < \theta_{th}. \end{cases} \tag{6}$$

$SE(z) = 1$ and $SE(z) = 0$ represent the SE pixel or non-SE pixel, respectively. In order to determine the value of θ_{th} more accurately, we set different thresholds to detect sharp edges in the standard SCV test sequences. As shown in Figure 4, when θ_{th} equals 800, the natural image area has more detailed information with lower contrast to be selected. When θ_{th} equals 1600 and 2000, partial edge of text is missing. Thus, sharp edges could be better filtered when θ_{th} is 1200.

3.2 | SE detection based SPR modeling for SCVs

Conventional spatial PR models were usually constructed considering multiple masking effects, such as the LA effect and the CM effect. Commonly, these masking effects are measured separately, the spatial PR threshold is then generated by nonlinear superposition of multiple estimated sub-PR thresholds.

The photoreceptor cells of HVS are able to adjust the signal conversion efficiency adaptively, which is usually quantified by

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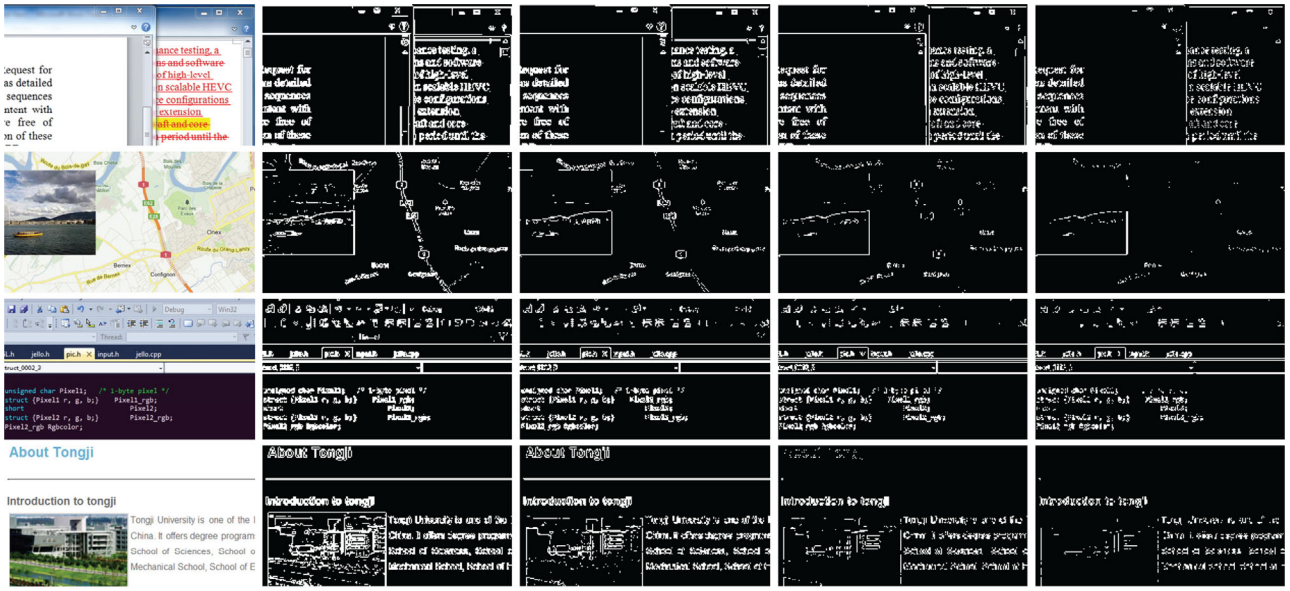


FIGURE 4 Typical samples in SCVs, and the sharp edge map under different thresholds

the LA effect, and visual sensitivity is relatively lower in both bright and dark areas. Thus, in [26], the sub-PR of LA effect is directly related to the background brightness. Specifically, the sub-PR J_{la} of pixel z could be estimated by:

$$J_{la}(z) = \begin{cases} \alpha_1 \times (1 - \sqrt{B(z)/127}) + \beta, B(z) \leq 127, \\ \alpha_2 \times \big(B(z) - 127\big) + \beta, B(z) > 127. \end{cases} \tag{7}$$

Constants α_1, α_2 and β are set to 17, 2/128, and 2 respectively. $B(z)$ represents the background luminance, which is calculated by the average luminance of the surrounding 5×5 block.

In terms of the CM effect, it represents the visibility reduction of one visual object in the presence of another. In the pixel domain, the sub-PR of CM effect is usually evaluated based on the gradient. Normally, the larger the gradient of the pixel, the higher the sub-PR of CM effect. When it comes to SCVs, they contain rich sharp edges whose gradient is always relatively larger. However, due to the selective response mechanism of visual neurons, the HVS is more sensitive to the sharp edges. In order to prevent erroneous estimates of masking effects in sharp edges, the SCVs will first be divided into the SE areas and the non-SE areas based on the SE detection, and their sub-PR of CM effects should be estimated with different approaches.

The distortion of sharp edge areas mainly appears in the form of the blur of the edge, and the pixel width of the sharp edge after blurring is about 3 [32]. So, we define the SE pixels and the surrounding 3×3 areas are the SE areas which are represented by $\mathbb{E} = SE \oplus B_1$. $\oplus$ is the image expansion operator and B_1 is the mean convolution template of 3×3 size. Two examples of $\mathbb{E}$ is shown in Figure 5, white areas and block areas in Figure 5b are the SE areas and the non-SE areas. The CM effects in SE areas and non-SE areas are called as the edge CM (ECM) effect and the texture CM (TCM) effect, respectively.

In vision science, the information in the visual system is represented according to contrast (i.e., relative changes in luminance). In other word, when contrast $\mathbb{C}$ is larger than the spatial PR ($J_S \leq \mathbb{C}$), the visual information could be perceived by HVS. In [32], Ferzli et al. proposed a concept of just noticeable blur (JNB). It was defined as the minimum value of blurriness that HVS can perceive around an edge, whose contrast is higher than the sub-PR of LA effect. Thus, in order to detect the edge, contrast should be larger than the sum of the JNB and the LA (i.e., $J_{la} + JNB \leq \mathbb{C}$). The blur of the edge is the main appearance of distortion of sharp edge areas, and JNB describes the visual characteristics of distortion at sharp edges, which is only related to edge contrast. This is consistent with the definition of contrast masking effect. As a result, we apply JNB to measure sub-PR of CM effect (J_{cm}) at sharp edges.

According to the analysis above, the sub-PR of CM effect for SE areas (J_{cm}^{SE}) could be estimated as follows:

$$J_{cm}^{SE}(z) = |I(z) - I_b(z)|. \tag{8}$$

Here, I_b is the blurred video in which the blurriness is just noticed. According to the literature [32], the blurred video is obtained through a Gaussian low-pass filter:

$$I_b(z) = I(z) * G(\sigma_{JNB}). \tag{9}$$

G represents a 7×7 Gaussian convolution kernel and its standard deviation is σ_{JNB} that makes the distortion reach the just noticeable point. σ_{JNB} is only related to the local contrast surrounding the target pixel z. According to Frtzli's vision experiment [32], the standard deviation of this gaussian low-pass filter could be approximately expressed by $\sigma_{JNB} = -\eta_1 \cdot \mathbb{C} + \eta_2$, η_1 and η_2 are set to 0.36 and 0.89 when the local contrast $\mathbb{C}$ is lower than 57, otherwise are set to 0.37 and 0.65, respectively.

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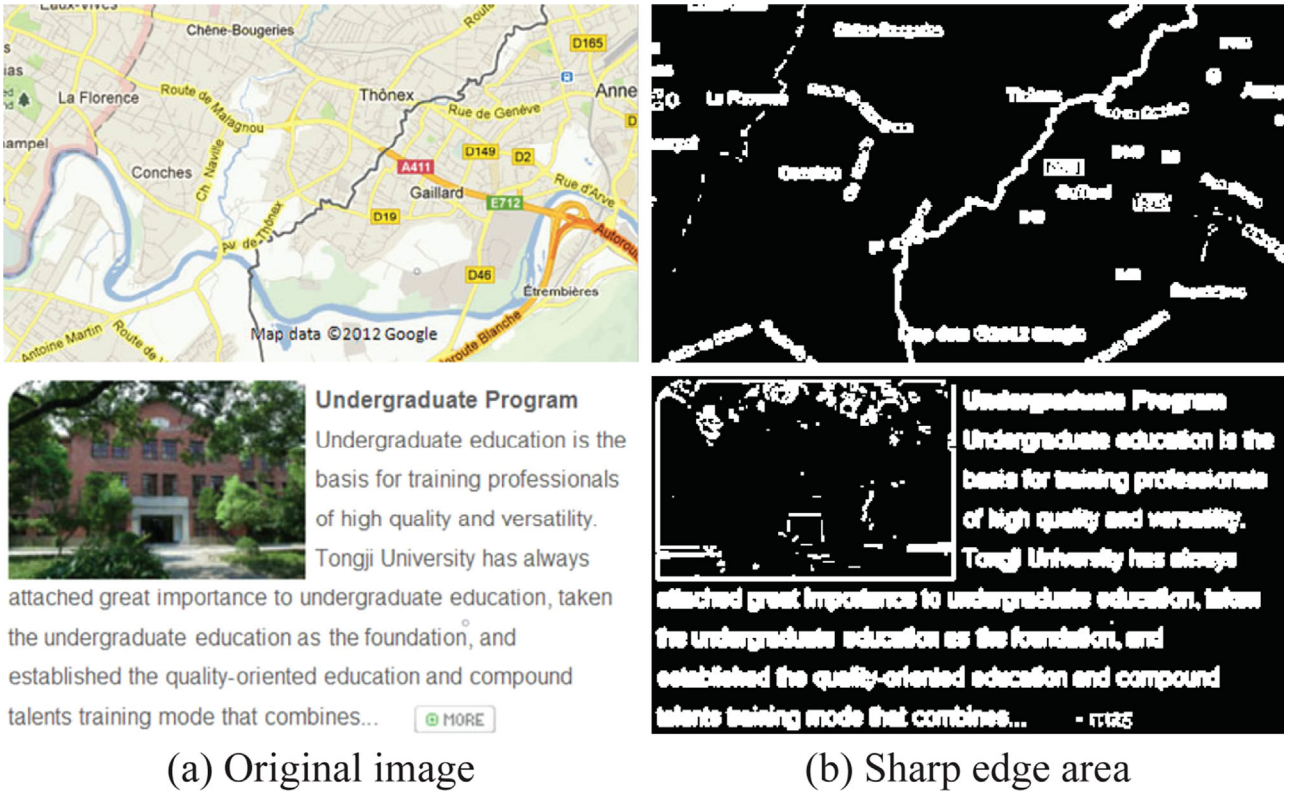


FIGURE 5 The selected sharp edge area of SCVs

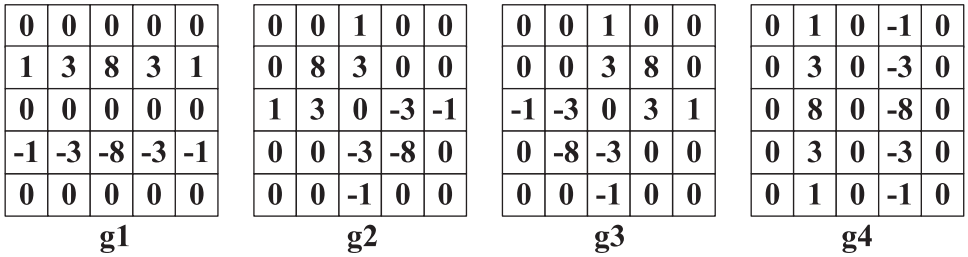


FIGURE 6 Four-direction high-pass filter for contrast masking

The contrast $\mathbb{C}$ is calculated by the difference between the maximum and minimum values in the surrounding 64×64 block.

For non-SE areas in SCVs, the sub-PR of CM effect is evaluated based on the maximum directional gradient of the pixel. In [33], the gradient of kth direction is expressed by

$$G_k(z) = \frac{1}{16} \sum_{i=1}^{5} \sum_{j=1}^{5} I(x-3+i, y-3+j) \times g_k(i,j), \quad (10)$$

where g_k represents the high-pass filter for k^{th} direction. Four directional high-pass filters are shown in Figure 6. The maximum directional gradient of the pixel z could be expressed with $G_m(z) = \max_{k=1,2,3,4}(G_k(z))$. Finally, the sub-PR of CM effect for non-SE areas could be calculated by

$$J_{cm}^{nSE}(z) = \left(0.01I(z) + 11.5\right)\left(0.01G_m(z) - 1\right) - 12. \quad (11)$$

For the pixel z, its sub-PR of CM effect could be estimated as follows according to the SE detection:

$$J_{cm}(z) = \begin{cases} J_{cm}^{SE}(z), & z \in \mathbb{E}(z), \\ J_{cm}^{nSE}, & z \notin \mathbb{E}(z). \end{cases} \quad (12)$$

In order to avoid the overestimation problem in SPR estimation, the spatial PR threshold is finally evaluated on the basis of the sub-PR of LA and CM effects through the nonlinear additivity

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model of masking (NAMM) [6].

$$J_S(z) = J_{la}(z) + J_{cm}(z) - C^{lc}\left(min\left(J_{la}(z), J_{cm}(z)\right)\right), \quad (13)$$

where C^{lc} is the gain reduction parameter and is set to 0.3 [6].

3.3 | Motion detection based temporal PR modeling for SCVs

In the temporal visual perception, the larger inter-frame luminance difference results into the larger visual masking effect. In [11], Chen et al. conducted the visual experiment and modeled the TPR as a scale factor for the SPR model. According to Chen's experiment, the inter-frame difference between t^{th} frame and $(t-1)^{\text{th}}$ frame is defined as follows:

$$R_T(z,t) = \frac{I(z,t) - I(z,t-1) + B(z,t) - B(z,t-1)}{2}. \quad (14)$$

Then, the conventional TPR model is built as the function of the inter-frame difference with a piecewise function as follows [11]:

$$J_{ct}(z,t) = max\left[\tau_1, \frac{\tau_2}{2} exp\left(\frac{-0.15}{2\pi}(255 - |R_T(z,t)|)\right) + \tau_1\right]. \quad (15)$$

τ_2 is set to 8 and 3.2 for $R_T(z,t) \le 0$ and $R_T(z,t) > 0$, respectively. τ_1 is set to 1.0. As described above, the conventional TPR model only considers the visual perception effect of inter-frame difference. Actually, human eye movement experiment has show that human eye always fixates at the foreground motion and ignores the background motion since the foreground motion usually provide a relatively strong surprise to the HVS [34]. Thus, the visual perception effect of inter-frame motion should be considered properly in the TPR modeling.

In order to improve the accuracy of TPR model, we propose a information-theoretic approach to measure the motion masking (MM) effect to the visual perception in the view of information theory. Specifically, the MM effect to the HVS is estimated with the self-information of the motion vector as follows:

$$\Gamma(\vec{v}) = -\log_2 p(\vec{v}). \quad (16)$$

As mentioned in Section 2.2, the motion information entropy of the frame H_p is calculated for frame category classification. $\Gamma(\vec{v})$ larger than H_p means that the HVS will pay more attention on the moving areas with the motion vector $\vec{v}$, which indicates the perceptual redundancy of these areas is relatively smaller. Thus, we set a motion scale factor for TPR model with the sigmoid function as follows:

$$\gamma_m(\vec{v}) = \gamma_0 + \frac{\gamma_1}{1 + e^{-\gamma_2(\Gamma(\vec{v}) - H_p(t))/H_p(t)}}. \quad (17)$$

γ_0, γ_1 and γ_2 are constants, which are set to 0.6, 0.8 and −2.0 according to our visual experiments. By the adjustment of the motion effect, the proposed TPR model is expressed by

$$J_T(z,t) = max\left[\tau_1, \frac{\gamma_m \cdot \tau_2}{2} exp\left(\frac{-0.15}{2\pi}(255 - |R_T(z,t)|)\right) + \tau_1\right]. \quad (18)$$

Figure 7 provides a comparison example between the conventional TPR model and the proposed TPR model. To visualize the motion between frames, we demonstrate the same part of the first two frames of 'FlyingGraphics'. Figures 7c and 7d are TPR maps generated by conventional TPR model and our TPR model, while brighter areas lead to larger TPR threshold. As shown in this figure, motion of lower proportion, such as local relative motion, will lead to a relatively smaller threshold in our TPR model, while TPR of motion of high proporion, such as background motion, has no significant changes.

4 | FOVEATED PR ESTIMATION FOR SCVs

In this section, we will first propose a set of the fixation point prediction scheme of SCVs. In particular, based on our previously proposed frame category classifier, fixation points of RM-frames and ST-frames are generated from the spatial-temporal saliency map of each frame, while fixation points of AB-frames are predicted from the positions of fixation points in several previous frames. The FPR model for SCVs is then modeled as a scale factor for spatial-temporal PR model according to the foveated masking effect.

4.1 | Fixation point prediction for SCVs

Spatial and temporal visual contents drives the movement of human eyes. Usually, during the saccade process, human eyes tend to find areas including interests and concerns, and fixate at the interesting areas to extract more visual details. According to the motion characteristics of SCVs, this paper classifies screen content frames into ST-frames, RM-frames and AB-frames. Thus, different schemes are designed to predict the coordinates of fixation points for three types' frames in this section.

Usually, the fixation points of images or videos could be predicted according to the saliency detection. In the ST-frame, there is no inter difference and inter motion among adjacent frames. In order word, the fixation points of ST-frames could be predicted according to the spatial saliency detection. In this paper, the detection of spatial saliency is based on the literature [35] and is represented by S_s. As for the RM-frame, we choose the normalized motion self-information to represent its

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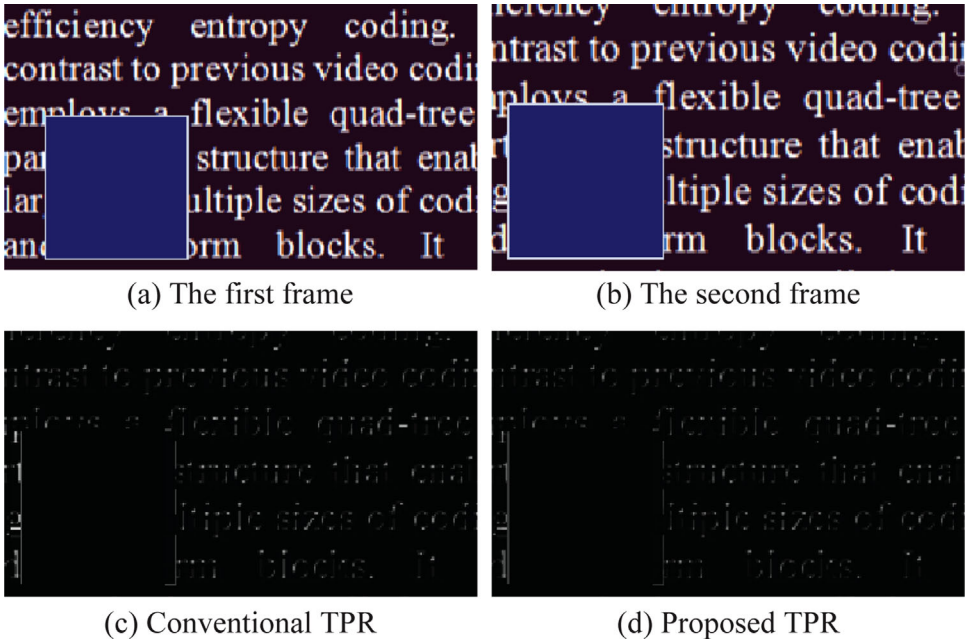


FIGURE 7 Comparison between the conventional TPR model and the proposed TPR model in part of 'FlyingGraphics'

temporal saliency S_t, which is calculated as follows:

$$S_t(\vec{v}) = \mathbb{N}(-\log_2 p(\vec{v})). \tag{19}$$

Then, the spatial-temporal saliency S_{st} of RM-frames is generated by fusing spatial saliency S_s and temporal saliency S_t through a parameterized normalization, sum and product (PNSP) method proposed in [36], calculated as (20). S_k represents spatial or temporal saliency map. γ_s, γ_t and γ_{st} are parameters to determine the weight of spatial and temporal saliency. We set $\gamma_{st} = (\gamma_s + \gamma_t)/2$, same as [36]. γ_k ($\gamma_k \in \gamma_s, \gamma_t$) is equal to the reciprocal of the spatial variance v_k of S_k, and can be calculated as (21). (x, y) indicates the pixel position, $S_{i,e}$ and $S_{j,e}$ are spatial expectation values in horizontal and vertical directions, obtained by (22).

$$S_{st} = \gamma_s S_s + \gamma_t S_t + \gamma_{st} S_s S_t, \tag{20}$$

$$v_k = \frac{\Sigma_{(x,y)} \sqrt{(x - S_{x,e})^2 + (y - S_{x,e})^2} \cdot S_k(x,y)}{\Sigma_{(x,y)} S_k(x,y)}, \tag{21}$$

$$S_{x,e} = \frac{\Sigma_{(x,y)} x \cdot S_k(x,y)}{\Sigma_{(x,y)} S_k(x,y)}, \quad S_{y,e} = \frac{\Sigma_{(x,y)} y \cdot S_k(x,y)}{\Sigma_{(x,y)} S_k(x,y)}. \tag{22}$$

Figure 8 shows the saliency maps of the 76th frame in 'WebBrowsing', which is an RM-frame with cursor movement. Spatial saliency map indicates that HVS pays more attention to text regions. Since HVS is sensitive to relative motion between frames, in the merged spatio-temporal saliency map, high saliency regions are concentrated on motion areas. In our previous work [37], when the maximal saliency in a local area is larger than 0.8, we found that there may be at least one fixation point in this area averagely. According to this observation, fixation areas are detected and the most salient points are defined as the predicted fixation points.

The predicted fixation points of non-AB-frames are shown in Figures 9a and 9b. As shown in the ST-frame, fixation points fall on titles, which are in line with the habit of browsing webpages (Figure 9a). In RM-frames, HVS is more sensitive to moving areas, such as cursor movement, pop-up windows etc., and the fixation point is located in these areas. Violent and irregular movements usually occur in AB-frames, resulting into the diffuse temporal salient areas as shown in Figure 10. Therefore, it is difficult to predict the location of fixation points in AB-frames based on spatial-temporal saliency detection. Actually, there is a short reaction time in the visual perception of HVS. In other words, the fixation points of AB-frames could be predicted according to the fixation points of previous frames.

As mentioned in Section 2.2, we have defined the visual reaction time t_{sr} and the perceptual sliding window l_{sw}. It is assumed that there is no other AB-frames in the l_{sw} before the current AB-frame. So, the movement of fixation points in the previous l_{sw} frames is static or approximately linear. Suppose that there are K fixation points in each frame and the coordinate of kth fixation points is (x_k, y_k). The coordinate predicted k^{th} fixation point in the tth frame could be predicted according to the fixation points in the sliding window l_{sw} as follows:

$$x_k(t) = x_k(t-1) + \frac{1}{l_{sw}} \sum_{i=2}^{l_{sw}+1} \left(x_k(t-i+1) - x_k(t-i)\right), \tag{23}$$

$$y_k(t) = y_k(t-1) + \frac{1}{l_{sw}} \sum_{i=2}^{l_{sw}+1} \left(y_k(t-i+1) - y_k(t-i)\right). \tag{24}$$

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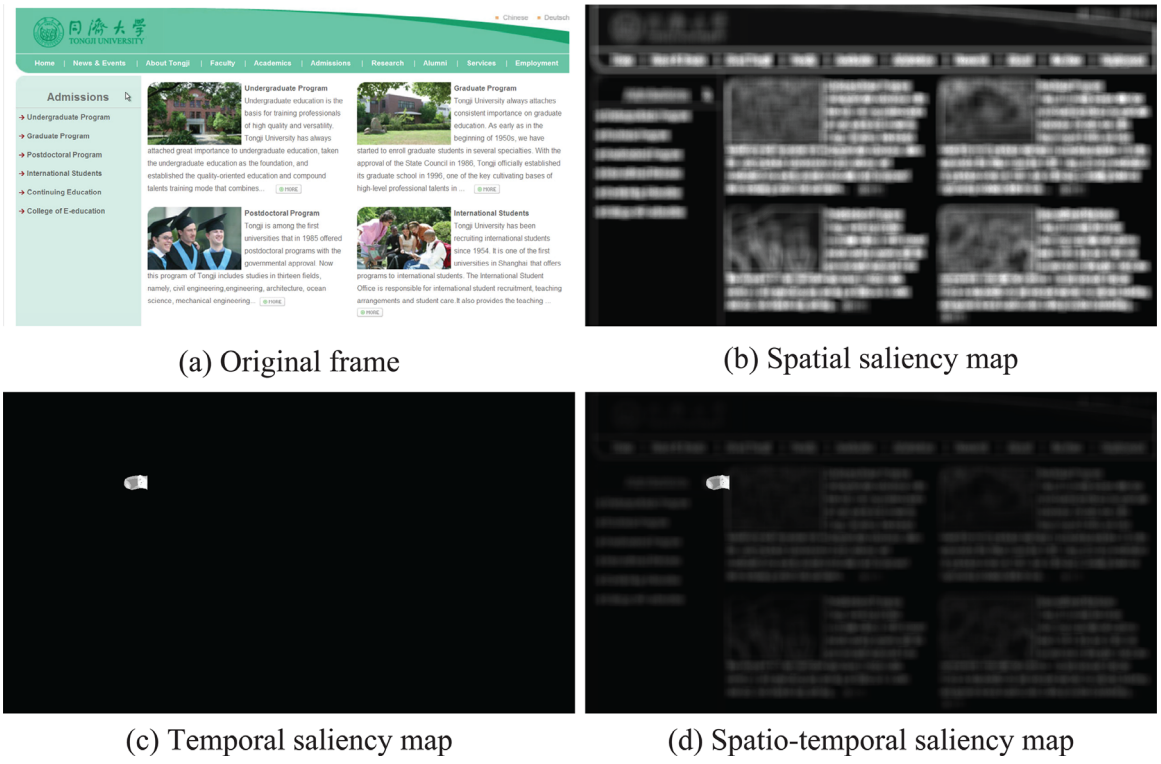


FIGURE 8 Saliency maps of the 76th frame in 'WebBrowsing'

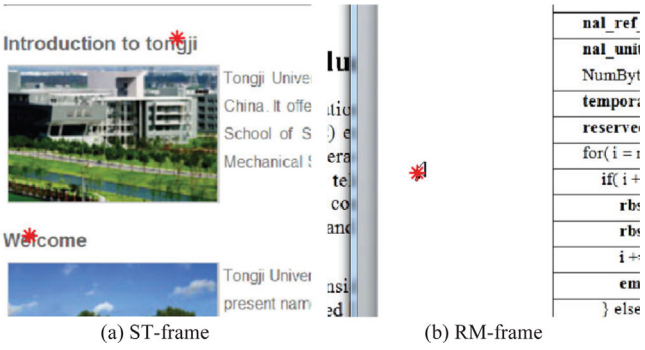


FIGURE 9 Predicted fixation points for non-AB-frames

As shown in Figure 11c,d, we provide two examples of the fixation points prediction in the AB-frame. To better show the scene switching between frames and the motion of fixation points, we demonstrate the previous frame corresponding to the AB-frame with fixation points in the slide window in Figure 11a,b, where the inter motion in the l_{sw} is the relative movement (cursor movement) and the static movement, respectively.

4.2 | Foveated PR model for screen content videos

Usually, the visual acuity decreases with the distance between the pixel and the fixation point. This visual characteristic is



FIGURE 10 Temporal saliency map of the 67th frame (AB-frame) in 'WebBrowsing'

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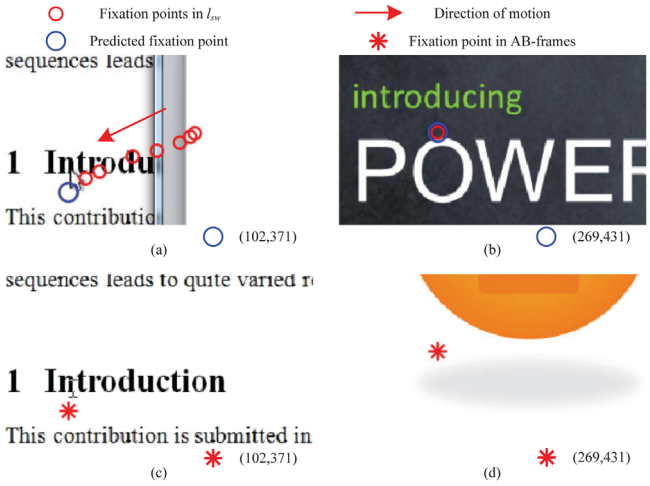


FIGURE 11 Predicted fixation points for AB-frames

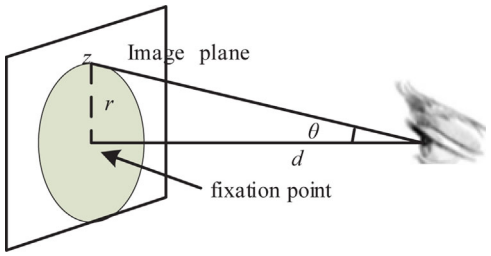


FIGURE 12 The retinal eccentricity

described by the foveated weighting model, and it is directly related to the retinal eccentricity. As shown in Figure 12, r is the distance between the pixel z and the fixation point. d is the viewing distance, and it is equal to 1.75 times than the hight of the screen. Thus, the retinal eccentricity θ could be represented with $arctan(r/d)$. According to the visual experimental results released in [38], Chen et al. presented the foveated weighting model as follows [11]:

$$W_f(\theta) = min\left(\frac{\theta_0}{\theta+\theta_0}, \frac{\pi d \varrho(\theta+\theta_0)}{360\cdot\theta_0 \times ln\left(\frac{1}{CT}\right)}\right). \quad (25)$$

In this model, CT, ϱ and θ_0 are model parameters, which represent the minimum contrast threshold, the spatial frequency decay constant and the half-resolution eccentricity constant respectively, and are set to 1/64, 0.106 and 2.3. Obviously, this model only reflects the mapping relation between the visual acuity and the retinal eccentricity. In our previous work [37], the visual attention was incorporated in FPR estimation. Assuming that there are K fixation points in the tth frame, considering the effect of visual attention, the visual acuity could be expressed by $\max_{k\in[1,\cdots,K]}[W_f(\theta_k)\cdot S_{ST}(z)]$ [37]. In addition, Chen et al. conducted the visual experiment and found that the perceptual redundancy not only relates to the retinal eccentricity, but also relates to the background luminance. Finally, the sub-PR of FM effect J_F is modeled as a scale factor as follows:

$$J_F(z) = \left(1 + (1 - \max_{k\in[1,\cdots,K]}[W_f(\theta_k)\cdot S_{ST}(z)])\right)^{\Omega(B(z))}, \quad (26)$$

$$\Omega(B(z)) = \mu_1 + \frac{1}{\sqrt{2\pi}\mu_2} exp\left(\frac{-(log_2(B(z)+1)-\mu_3)^2}{2\sigma^2}\right). \quad (27)$$

Here, μ_1, μ_2 and μ_3 are model parameters and set to 0.5, 0.8 and 7 respectively. Finally, the SCV-PR model is represented as the product of SPR, TPR and FPR, that is

$$J(z) = J_S(z) \times J_T(z) \times J_F(z). \quad (28)$$

5 | EXPERIMENTS AND RESULTS

In this section, the accuracy of the proposed SCV-based PR model is evaluated and compared with four state-of-the-art PR models. Then, the proposed PR model will be incorporated into SCV compression to evaluate its effectiveness.

5.1 | The accuracy of the proposed PR model

Normally, the accuracy of PR model could be evaluated by the ability of hiding noise. More specifically, the noise should be injected into the original video under the guidance of the PR model as follows:

$$\hat{I}(z) = I(z) + \zeta \cdot S_r(z) \cdot J(z). \quad (29)$$

Here, I and $\hat{I}$ represent the original video and the noised video. $S_r(\cdot)$ represents the bipolar random noise of ± 1. ζ is the noise energy adjustment factor. By adjusting the value of this factor, different PR models guide the injected noise with similar noise energy. Usually, the noise in the noised video is measured by peak signal to noise ratio (PSNR). The perceptual quality of noised video is measured by the differential mean opinion score (DMOS). The smaller the PSNR of noised video, and the closer the DMOS is to zero, indicating that the corresponding PR model is more accurate.

The subjective quality assessment experiment is performed according to the double stimulus continuous quality scale (DSCQS) method [39]. The presentation strategy of DSCQS is shown in Figure 14, in which A and B are the original video and the noised video. In our experiment, both videos are presented pseudo-randomly on a 17-in monitor and 14 subjects (7 males and 7 females) are asked to give the subjective quality scores of both videos during the voting time. The viewing distance of audiences is set to 1.75 times of the screen height. The range of

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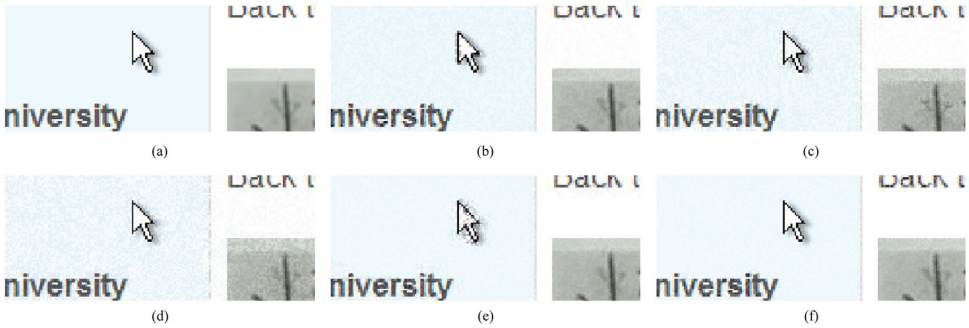


FIGURE 13 Visual quality comparison of the noised fixation area generated by PR models in 'WebBrowsing'. (a) Original video; (b) Wang's model [26] (L-PSNR=30.766); (c) Wu's model [9] (L-PSNR=32.703); (d) Zeng's model [10] (L-PSNR=31.139); (e) Wang's model [28] (L-PSNR=30.412); (f) Proposed model(L-PSNR=34.137)

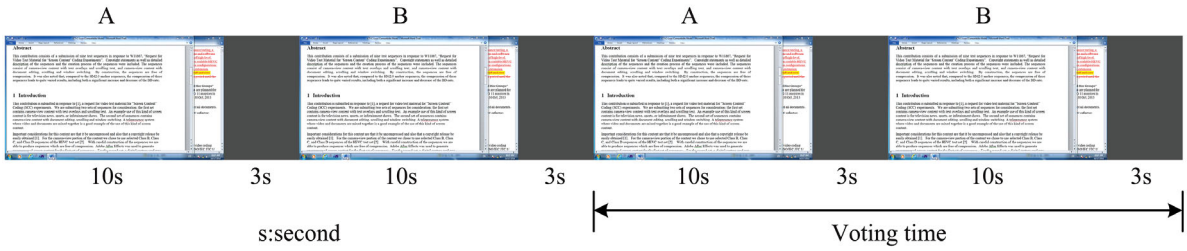


FIGURE 14 Presentation strategy of DSCQS

subjective quality score (SQS) is shown in Table 3. The DMOS value between the original video and the noised video is calculated as follows:

$$DMOS = \sum_{k=1}^{14} SQS_J(k)/14 - \sum_{k=1}^{14} SQS_O(k)/14, \qquad (30)$$

where SQS_J and SQS_O are the subjective quality score of the noised video and the original video, respectively. k is the subject index. As the DMOS equation shows, the DMOS is closer to zero, which means that the noised video is more similar with the original video.

In order to intuitively reflect the accuracy of the proposed PR model, we compare it with four state-of-the-art PR models which are named as 'Wang2016' [26], 'Wu2017' [9], 'Zeng2019' [10] and 'Wang2021' [28] in this paper. Wu2017 and Zeng2019 are two pixel-based PR models proposed for natural videos. In [9], Wu et al. proposed a pattern masking effect instead of the CM effect to improve the accuracy the sub-PR model of CM effect. In [10], Zeng et al. divided the CM effect into the edge masking, the orderly texture masking effect and the disorderly texture masking effect. Wang2016 and Wang2021 are two PR models designed for screen content images. In [26], the PR model was built only considering the LA effect and the CM effect. In [28], Wang et al. estimated high-frequency and orientation sensitivity of HVS when perceiving SCIs besides LA and CM effect. These PR models are tested in the standard sequences [40] and the detailed results of all sequences are shown in Table 2. As a reference, we also present the average SSIM for each model in the table.

As shown in this table, the average PSNRs of five PR models are 29.035 dB, 30.583 dB, 29.564 dB, 28.938 dB and 28.870 dB, respectively. Obviously, the average PSNR of the proposed PR model is smallest, indicating that more noise is added into the original video. The average SSIMs of Wu2017, Wang2021 and the proposed model are comparatively close, indicating that in terms of structural similarity, these models perform almost the same, while SSIMs of Wang2016 and Zeng2019 are relatively small. The average DMOS values of these PR models are −4.01, −5.70, −3.96, −3.84 and −2.96. The proposed PR model has DMOS closest to zero compared to other PR models. Under the guidance of the proposed PR model, SCVs could be injected more noise with better perceptual quality. In order to more intuitively compare the perceptual quality of noised SCVs with different PR models, we have provided two examples shown in the Figures 13 and 15.

As shown in Figure 13a, this picture is an original video of the 39th frame in the sequence 'WebBrowsing'. Other pictures in this figure are the noised video by different PR models. Notice that the noise injected into the whole frame is almost the same for different models. This frame is an RM-frame with cursor movement, which means the area in Figure 13 is a fixation area, and HVS can tolerate less distortion. To objectively illustrate the noise injected into the area, we list the local PSNR (L-PSNR) of the noised pictures. It can be found that there is almost no significant difference between Figures 13a and 13f with the highest local PSNR, indicating that the proposed model performs best in maintaining quality of fixation areas. In contrast, the noise in Figure 13b–e could be perceived by the HVS in the background, which means that the PR threshold of the background

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TABLE 2 Model performance comparisons

	Wang2016			Wu2017			Zeng2019			Wang2021			Proposed		
Videos	**PSNR(dB)**	**SSIM**	**DMOS**	**PSNR(dB)**	**SSIM**	**DMOS**	**PSNR(dB)**	**SSIM**	**DMOS**	**PSNR(dB)**	**SSIM**	**DMOS**	**PSNR(dB)**	**SSIM**	**DMOS**
Map	32.304	0.950	−3.21	30.650	0.891	−8.07	31.239	0.861	−6.64	31.528	0.924	−3.21	31.073	0.944	−2.43
Programming	27.579	0.885	−3.43	29.949	0.917	−6.14	28.854	0.867	−4.07	27.178	0.914	−3.00	27.964	0.919	−2.86
Robot	31.459	0.795	−4.29	30.097	0.806	−6.71	29.532	0.735	−4.14	31.312	0.806	−4.07	31.249	0.789	−3.00
sc_console	25.759	0.498	−3.50	28.975	0.623	−5.57	25.805	0.499	−4.64	28.374	0.630	−3.57	24.663	0.559	−3.21
sc_desktop	27.712	0.836	−5.43	28.978	0.875	−8.14	28.256	0.832	−4.14	26.255	0.877	−5.00	27.636	0.880	−4.86
sc_flyingGraphics	28.429	0.934	−3.57	30.687	0.945	−3.07	30.553	0.923	−2.71	30.081	0.958	−3.93	28.484	0.953	−2.07
SlideShow	29.740	0.670	−3.21	33.171	0.724	−3.93	28.896	0.621	−3.07	29.663	0.622	−3.29	29.488	0.641	−2.86
WebBrowsing	31.288	0.941	−4.21	31.775	0.937	−4.93	31.980	0.902	−3.79	32.800	0.976	−3.71	30.611	0.963	−3.21
WordEditing	27.048	0.918	−5.21	30.959	0.921	−4.71	30.957	0.877	−2.43	28.839	0.962	−4.86	28.659	0.947	−2.14
Average	**29.035**	**0.825**	**−4.01**	**30.583**	**0.849**	**−5.70**	**29.564**	**0.791**	**−3.96**	**29.559**	**0.852**	**−3.85**	**28.870**	**0.844**	**−2.96**

TABLE 3 Evaluation criteria of the subjective experiment

SQS	**0–20**	**20–40**	**40–60**	**60–80**	**80–100**
Quality	Bad	Poor	Fair	Good	Excellent

is overestimated by these PR models. Figure 15 shows the text area of the first frame of the sequence 'WordEditing'. Again, the noise energy injected into the whole frame with different models is almost the same. It could be found that the proposed PR model could better maintain the structure of edges, as well as higher subjective quality of the background. In summary, the experimental result has shown that the proposed PR model outperforms the existing PR models in terms of accuracy measurement.

In addition, the proposed PR model consists of the spatial PR model, the temporal PR model and the foveated PR model. The TPR model and the FPR model are designed as two scale factors for the SPR model. In order to show the accuracy of three models, we have provided the noised videos guided by the proposed SPR model, the proposed STPR model and the STFPR model as shown in Figure 16. As shown in this figure, the region in the first row has no inter-frame variation, so the local PSNR(L-PSNR) of SPR and STPR model are the same. When there exist inter-frame changes, the PSNR of SPR model is smallest and that of STFPR model is highest, which means that more noise is injected into the original video guided by the TPR model and the FPR model. However, the perceptual qualities of three noised videos are almost the same as that of the original video. So, introducing the TPR model and FPR model could improve the accuracy of PR estimation for SCVs.

5.2 | Performance of the proposed PR model

In order to evaluate the effectiveness of the proposed PR in SCV compression, we have incorporated the proposed PR model and other four comparison PR models into the HEVC-SCC reference software HM-16.20+SCM-8.8 [41]. Specifically, during the compression process of SCVs, the quantization parameter (QP) of each coding unit (CU) is readjusted based on PR maps to reduce the perceptual redundancy, so that the compression efficiency is further improved. Since HVS can tolerate more distortion in regions with higher PR threshold, higher QP is set to these areas to save more bits. In this article, we adopt the QP adjustment algorithm proposed in [11]. The CU-level perceptual QP (QP_{CU}) is obtained by (31).

$$QP_{CU} = \sqrt{\frac{1}{\varphi}} QP_f, \tag{31}$$

where QP_f is the QP level of the current frame, φ is a parameter related to the average PR threshold of the current CU $\overline{J_{CU}}$ and the average PR intensity of the whole frame $\overline{J_F}$, calculated as (32). χ_0, χ_1, χ_2 are constants which are set to 0.7, 0.6 and 4.0

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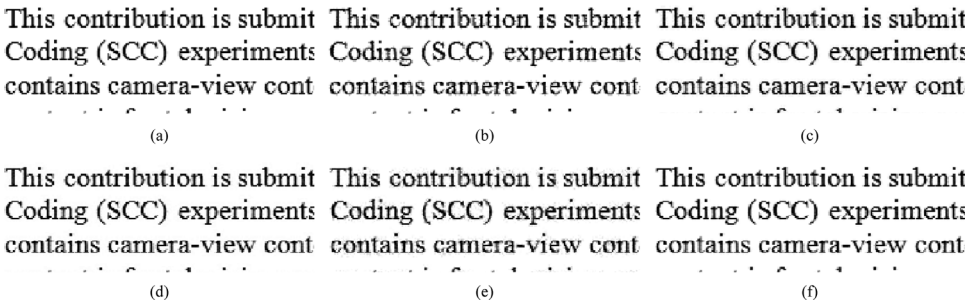


FIGURE 15 Visual quality comparison of the noised text area generated by PR models in 'WordEditing'. (a) Original video; (b) Wang's model [26] (L-PSNR=25.530); (c) Wu's model [9] (L-PSNR=29.306); (d) Zeng's model [10] (L-PSNR=29.793); (e) Wang's model [28] (L-PSNR=24.366); (f) Proposed model(L-PSNR=30.829)

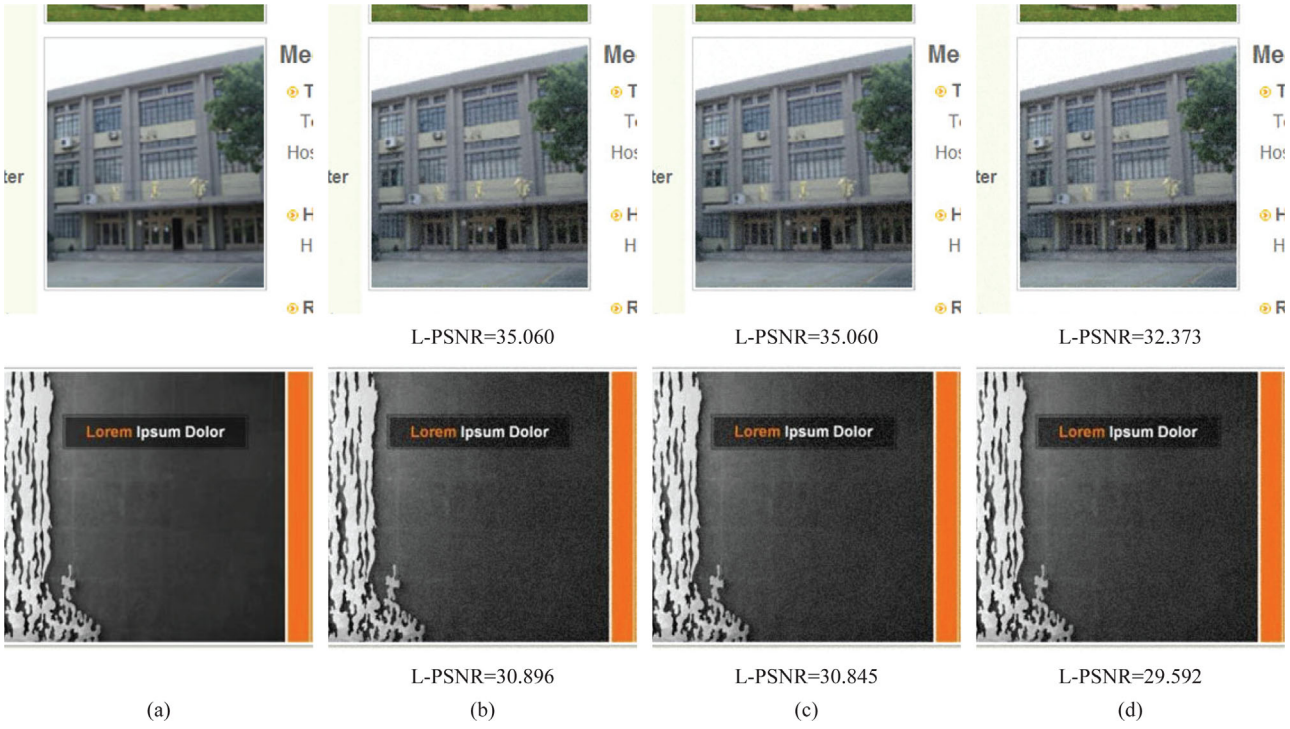


FIGURE 16 Visual quality comparison of the noised SCV generated by different components in the proposed PR model. (a) Original video; (b) Proposed SPR model; (c) Proposed STPR model; (d) Proposed STFPR model

respectively.

$$\varphi = \chi_0 + \frac{\chi_1}{1 + \exp(-\chi_2(\overline{J_{CU}} - \overline{J_F})/\overline{J_F})}. \tag{32}$$

Nine sequences from common test conditions for screen content coding [40] are compressed with four QPs (22, 27, 32, 37) under random-access(RA) main profile. Similar with ζ in (29), which ensures that different PR models are able to share similar PR energy, here, we adjust the energy of each models to ensure that similar bit rate savings are achieved. The subjective quality of compressed video is measured by the DMOS and the subjective quality assessment experiment is also performed according to the DSCQS method. In this experiment, A and B videos are the reconstructed videos which are compressed by the original encoder and the perceptual encoder, respectively. Table 4 shows the ratio of bitrate reduction (ΔR), the loss of objective quality ($\Delta PSNR$) and the loss of subjective quality (DMOS). ΔR and $\Delta PSNR$ are calculated wtih $\Delta R = 100 \times (R_p - R_o)/R_o$, $\Delta PSNR = (PSNR_p - PSNR_o)$, while '$p$' and '$o$' represents perceptual SCV compression and original SCV compression, respectively. The retailed experimental result is shown in Table 4. From this table, we can found that the average bitrate reductions of five PR models are very close, which are in the range of [−7.80, −7.26]. The average PSNRs of perceptual compressed videos approximately decrease by 0.93 dB. Besides, the average DMOS values of five PR models are −0.31, −0.48, −0.33, −0.29 and −0.12, which are close

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TABLE 4 Test results of perceptual SCV compression with different PR models

		Wang2016 [26]			Wu2017 [9]			Zeng2019 [10]			Wang2021 [28]			Proposed		
Videos	**QP**	**ΔR(%)**	**ΔPSNR**	**DMOS**	**ΔR(%)**	**ΔPSNR**	**DMOS**	**ΔR(%)**	**ΔPSNR**	**DMOS**	**ΔR(%)**	**ΔPSNR**	**DMOS**	**ΔR(%)**	**ΔPSNR**	**DMOS**
Console	22	−5.66	−1.046	−0.3	−5.73	−1.104	−0.1	−6.03	−1.108	−0.1	−5.93	−1.106	−0.1	−6.17	−1.190	−0.2
	27	−5.35	−1.045	−0.1	−5.72	−1.125	−0.3	−5.41	−1.096	−0.1	−5.31	−1.017	−0.2	−5.56	−1.153	−0.1
	32	−5.43	−0.965	−0.1	−5.67	−0.985	−0.2	−6.05	−0.977	−0.2	−5.29	−0.942	−0.1	−5.72	−1.048	0.0
	37	−5.84	−0.867	−0.2	−5.84	−0.799	−0.1	−6.06	−0.811	−0.1	−5.62	−0.832	−0.2	−6.25	−0.941	−0.1
Desktop	22	−3.77	−1.222	−0.2	−4.04	−1.453	−0.6	−3.90	−1.249	−0.3	−3.80	−1.288	−0.1	−3.56	−1.140	0.1
	27	−4.00	−2.259	−0.1	−4.20	−2.419	−0.4	−4.14	−2.237	−0.6	−3.57	−2.219	−0.2	−3.80	−2.261	0.0
	32	−3.87	−2.206	−0.3	−4.14	−2.282	−0.5	−4.01	−2.350	−0.3	−4.23	−2.344	−0.4	−3.25	−2.045	−0.1
	37	−7.68	−2.628	−0.4	−7.10	−2.251	−0.3	−7.36	−2.533	−0.1	−7.07	−2.525	−0.3	−6.78	−2.289	−0.3
FlyingGraphics	22	−11.66	−1.497	−0.4	−12.52	−1.515	−0.2	−12.12	−1.529	−0.3	−12.27	−1.504	−0.3	−11.59	−1.507	−0.1
	27	−9.17	−0.909	−0.3	−9.81	−0.900	−0.5	−9.29	−0.873	−0.4	−9.24	−0.911	−0.4	−9.27	−0.913	−0.2
	32	−8.19	−0.736	−0.1	−9.24	−0.823	−0.4	−8.39	−0.708	−0.2	−8.09	−0.707	−0.2	−8.22	−0.748	0.0
	37	−9.55	−0.809	−0.4	−10.48	−0.816	−0.2	−9.58	−0.754	−0.1	−9.53	−0.803	−0.1	−9.76	−0.851	−0.1
Map	22	−5.32	−0.855	−0.1	−5.71	−0.928	−1.4	−5.15	−0.837	−0.3	−5.38	−0.907	−0.2	−5.44	−0.907	−0.1
	27	−7.56	−1.044	−0.1	−7.86	−1.007	−1.2	−6.77	−0.899	−0.6	−7.40	−1.024	−0.1	−7.42	−0.970	−0.4
	32	−7.88	−0.725	−0.2	−8.38	−0.813	−1.3	−7.90	−0.731	−0.4	−8.01	−0.731	−0.1	−8.16	−0.791	−0.1
	37	−6.54	−0.494	0.0	−7.04	−0.476	−1.5	−7.18	−0.476	−0.4	−6.17	−0.484	−0.2	−7.40	−0.511	−0.2
Programming	22	−7.34	−0.660	−0.2	−9.43	−0.734	−0.5	−7.70	−0.576	−0.3	−7.59	−0.559	−0.1	−7.52	−0.673	0.1
	27	−6.48	−0.768	0.1	−8.07	−0.813	−0.4	−6.11	−0.613	−0.4	−5.67	−0.602	−0.1	−6.75	−0.796	−0.1
	32	−6.37	−0.688	0.0	−7.24	−0.672	−0.3	−5.97	−0.579	−0.2	−6.67	−0.682	−0.1	−5.95	−0.673	0.0
	37	−5.21	−0.720	−0.2	−6.10	−0.745	−0.4	−4.89	−0.676	−0.4	−4.09	−0.627	−0.2	−5.36	−0.765	−0.2
Robot	22	−16.44	−0.571	−1.1	−16.86	−0.611	−0.1	−16.78	−0.604	0.0	−16.71	−0.603	−0.9	−16.34	−0.569	−0.6
	27	−15.01	−0.493	−1.3	−15.29	−0.518	0.1	−15.54	−0.457	−0.2	−15.15	−0.486	−0.7	−14.85	−0.491	−0.5
	32	−13.95	−0.408	−1.0	−14.42	−0.412	−0.2	−14.80	−0.389	−0.1	−14.13	−0.403	−0.6	−13.64	−0.412	−0.2
	37	−11.97	−0.377	−1.1	−13.14	−0.386	−0.1	−12.69	−0.349	−0.2	−12.53	−0.383	−0.6	−12.13	−0.396	−0.4
SlideShow	22	−8.36	−0.474	−0.4	−9.24	−0.486	−0.6	−8.83	−0.470	−0.4	−8.80	−0.475	−0.4	−9.60	−0.486	0.0
	27	−8.21	−0.542	−0.2	−8.79	−0.537	−0.5	−7.94	−0.529	−0.3	−6.90	−0.524	−0.4	−8.48	−0.544	0.0
	32	−6.15	−0.689	−0.2	−6.96	−0.609	−0.6	−6.31	−0.749	−0.1	−4.71	−0.659	−0.5	−6.30	−0.697	−0.2
	37	−5.80	−0.628	−0.4	−6.43	−0.644	−0.7	−4.85	−0.527	−0.4	−6.06	−0.563	−0.3	−4.42	−0.405	−0.1

(Continues)

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TABLE 4 (Continued)

		Wang2016 [26]			Wu2017 [9]			Zeng2019 [10]			Wang2021 [28]			Proposed		
Videos	**QP**	**ΔR(%)**	**ΔPSNR**	**DMOS**	**ΔR(%)**	**ΔPSNR**	**DMOS**	**ΔR(%)**	**ΔPSNR**	**DMOS**	**ΔR(%)**	**ΔPSNR**	**DMOS**	**ΔR(%)**	**ΔPSNR**	**DMOS**
WebBrowsing	22	−5.37	−0.883	−0.3	−5.46	−0.894	−0.7	−5.60	−0.860	−0.6	−5.71	−0.899	−0.3	−5.36	−0.897	0.1
	27	−4.83	−1.116	−0.5	−4.76	−1.083	−0.3	−4.70	−1.035	−0.9	−4.44	−1.115	−0.4	−5.07	−1.179	0.2
	32	−3.08	−1.066	−0.2	−3.05	−1.083	−0.4	−3.34	−0.932	−0.5	−3.94	−1.062	−0.4	−3.00	−0.914	−0.1
	37	−4.00	−0.836	−0.1	−3.51	−1.161	−0.6	−3.21	−1.032	−0.4	−4.06	−1.109	−0.1	−3.73	−1.231	0.0
WordEditing	22	−6.57	−0.818	−0.2	−8.46	−0.798	−0.4	−8.82	−0.815	−0.5	−7.01	−0.809	−0.1	−9.09	−0.821	0.1
	27	−6.09	−0.705	−0.4	−7.48	−0.794	−0.3	−8.16	−0.807	−0.4	−6.38	−0.736	−0.3	−7.85	−0.801	0.0
	32	−5.23	−0.852	−0.1	−5.55	−0.636	−0.4	−6.29	−0.645	−0.7	−5.52	−0.862	−0.1	−5.91	−0.768	−0.2
	37	−7.51	−0.844	−0.1	−7.16	−0.701	−0.6	−8.07	−0.718	−0.6	−6.87	−0.823	−0.1	−7.42	−0.827	−0.1
Average		−7.26	−0.929	−0.31	−7.80	−0.945	−0.48	−7.50	−0.904	−0.33	−7.22	−0.926	−0.28	−7.42	−0.934	−0.12

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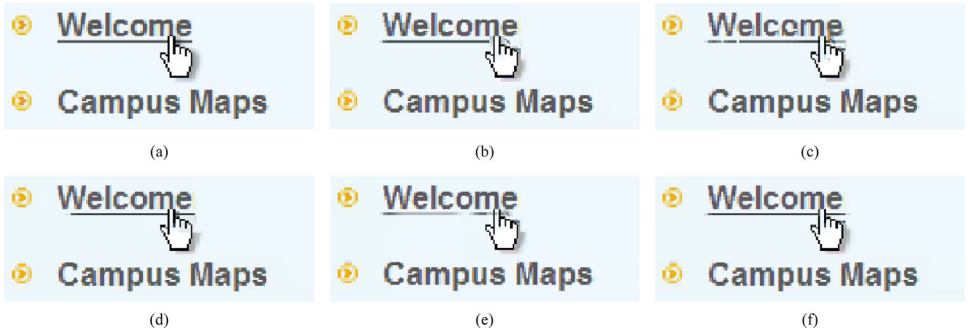


FIGURE 17 Visual quality comparison of the distorted text area generated by different PR models in 'WordEditing'. (a) Original video; (b) Wang's model [26] (L-PSNR=30.420); (c) Wu's model [9] (L-PSNR=29.198); (d) Zeng's model [10] (L-PSNR=30.062); (e) Wang's model [28] (L-PSNR=28.133); (f) Proposed model(L-PSNR=31.042)

to zero. However, the DMOS of the proposed PR model is smallest, showing that the perceptual quality of compressed SCVs is best under the guidance of the proposed PR model.

Figure 17 demonstrates part of the 14th frame in the sequence 'WebBrowsing' to intuitively compare the perceptual compression results with different PR models. As shown in this figure, this area contains a cursor with relative motion. In the visual perception, the HVS is sensitive to this area and the coding distortion is easy to be perceived. From this figure, we can find that our model performs best on maintaining the integrity of lines and structures. In general, the proposed PR model is more accurate than existing PR models and performs better in SCV perceptual compression.

5.3 | Discussion and future work

The proposed perceptual redundancy model is mainly applied in the process of perceptual video coding of screen content, and we adopted a perceptual video coding strategy designed for natural videos. In the next step of our research, unique encoding features of HEVC-SCC will be taken into account for further improvement in bit rate savings. The proposed method does not include any deep learning-based approach. Subsequently, we will work on constructing datasets suitable for PR model for SCVs perception, and we will develop a learning-based model for compression of SCVs.

Furthermore, HVS cannot perceive subtle differences between compressed videos when QPs are relatively close, this means that we can cluster compressed videos into several layers according to the QP. For each layer, the perceived quality can be considered roughly equal. Then, according to the perception model and the layers, the most appropriate QP is assigned to each area of the video, reducing the bit rate substantially without degradation in subjective quality.

6 | CONCLUSION

In this paper, a novel perceptual redundancy model has been proposed for screen content videos in pixel domain. This model consists of spatial PR, temporal PR and foveated PR. Because of the sharp edges of SCVs, this paper detected the sharp edge areas and estimated the spatial perceptual redundancy of these areas based on a perceptual algorithm of edge blur. The inter-motion on visual perception was analyzed and the motion masking effect is first incorporated into TPR modeling in this paper. In order to predict the fixation points more accurately, each frame of SCVs was classified based on motion detection and different methods were used to predict fixation points in different types of frames. Then, the FPR model was built considering the foveated masking effect and the visual attention. Finally, the perceptual redundancy of SCV was estimated according to the product of spatial PR, temporal PR and foveated PR. Numerous experiments indicates that compared to state-of-the-art PR models, the proposed model performs best in accuracy evaluation and effectiveness evaluation. Moreover, we incorporate our model into SCV compression with an adaptive perceptual quantizer, and an average bit saving of 7.42% can be achieved with less decline in subjective quality.

ACKNOWLEDGEMENTS

This work was supported in part by the NSFC under Grant 61871437; in part by the Natural Science Foundation of Hubei Province of China under Grant 2019CFA022. J.L. and L.Y. are with the College of Electronic Information and Communications, Huazhong University of Science and Technology, Wuhan, China. H.W. is with School of Communication Engineering, Hangzhou Dianzi University, Hangzhou, China.

The authors would also like to thank the editors and anonymous reviewers for their insightful comments, which have greatly improved the quality of this paper.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Research data are not shared.

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How to cite this article: Li, J., Yu, L., Wang, H.: Perceptual redundancy model for compression of screen content videos. IET Image Process. 16, 1724–1741 (2022). https://doi.org/10.1049/ipr2.12443

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